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ADSORPTION MECHANISM OF POLYLYSINE ON HYDROXYAPATITE AND ITS EFFECT ON DISSOLUTION PROPERTIES OF HYDROXYAPATITE

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Summary—Adsorbed amounts of poly-L-lysine (pLys) and bromide ion on hydroxyapatite (HAp) from aqueous solutions of poly-L-lysine hydrobromide, and amounts of calcium and phosphate ions liberated concurrently from HAp during the adsorption of pLys were determined at 25°. The pLys was adsorbed on HAp by the mechanism of ion-exchange between its amino groups and calcium ions of HAp. The released amount of calcium ion increased, therefore, with the adsorbed amount of pLys. On the other hand, the released amount of phosphate ion first decreased and then increased after attaining a minimum with the equilibrium concentration of pLys. The analysis using an equilibrium dialysis method revealed that the released phosphate ions were mainly in the bound state to the amino groups of pLys remaining in the solution, and that the concentrations of calcium and phosphate ions free from both HAp and pLys were restricted by each other under the law of the solubility product of HAp. The first decrease in the released amount of phosphate ion was concluded to be attributed primarily to the increase in the released amount of calcium ion because pLys remaining in the solution was little in this region. When sodium hydroxide was added to the solution, the adsorbed amount of pLys increased and then slightly decreased with the equilibrium pH of the solution due to the increase or decrease of the electrostatic attractive force between the adsorbate and the adsorbent. However, conformational change in pLys around pH 10 seemed to have little effect on the adsorption.

Hydroxyapatite (HAp), a basic calcium phosphate which has the chemical formula $Ca_{10}(PO_4)_6(OH)_2$, is the most stable phase of calcium phosphates under ordinary conditions.[1,2] This has been widely used for chromatographic separations of proteins, nucleic acids, enzymes, and so on, since Tiselius *et al.* introduced it as a packing material for liquid chromatography in 1950's.[3] Today, many commercial HAp-columns for high performance liquid chromatography are on the market.

In the studies on HAp-column chromatography, the factors governing interactions between proteins and HAp have been investigated, and some mechanisms, such as specific or non-specific complex formation between functional groups of protein and calcium or phosphate ions of HAp and as ion-exchange between the groups and the ions, were proposed.[4–7] However, detailed mechanism of the adsorption is still unknown because complexity of conformation and diversity of side chains of proteins make the elucidation of the mechanism difficult.

On the other hand, HAp is known to dissolve slightly in aqueous media. This behaviour is complicated and, in spite of many studies,[8–12] not elucidated entirely, mainly because HAp exists in the form of very small crystallites and its physicochemical properties depend on the surface state.[13] The dissolution properties of HAp seem important in connection with the durability of HAp-columns.

Poly-L-lysine (pLys) is a typical homopolypeptide, the molecular structure of which has been studied extensively. In the present study, the interactions between pLys and HAp are investigated by using an equilibrium dialysis technique. The adsorbed amount of pLys on HAp and the amounts of calcium and phosphate ions liberated from HAp are determined, and the results are discussed taking into account ion-exchange between calcium ions of HAp and amino groups of adsorbed pLys, binding of phosphate ions to the groups of pLys in solution, and the solubility product of HAp.

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EXPERIMENTAL

Reagents

Hydroxyapatite was synthesized according to the method of Avnimelech *et al.*[14] The calcium/phosphorus ratio was 10.38/6, and the point of zero charge (PZC) determined by pH titration was 6.70. The specific surface area determined by the BET procedure at 77 K with nitrogen gas as an adsorbate was 29.0 m^2/g. Poly-L-lysine hydrobromide (pLysHBr) was purchased from Sigma Chemical Co., and its viscosity average molecular weight assessed by the supplier was 21,500. The standardization of a stock solution of the pLysHBr was performed by the colloidal titration method[15] for amino group of pLys, using a standard solution of potassium polyvinyl sulphate (Wako Pure Chemical Industries, Ltd.) with Toluidine Blue indicator, and by the mercuric thiocyanate method[16,17] for bromide ion. Water was demineralized with a Milli-QSP water purification system (Millipore Co.).

Apparatus

A Nippon Jarrel-Ash ICAP-750N multichannel inductively coupled plasma atomic emission spectrometer was used. The nebulizer used was a conventional cross-flow type and argon gas flow-rates were 0.5 (carrier), 0.6 (auxiliary) and 16 (coolant) l l./min. A Hitachi 330 spectrophotometer, Hitachi 650 spectrofluorometer, Jasco J-600 spectropolarimeter and Horiba F-11 pH meter were also used for the analyses described below.

Procedure

The adsorption experiment was carried out by suspending HAp (1 g) to aqueous solutions of pLysHBr (50 ml) at 25° for 5 days. After filtration with a Millipore filter with a pore size of 0.1 μm, the filtrates were analysed.

The dialysis experiment was carried out as follows: HAp (0.6 g) was suspended in pLysHBr solutions (30 ml). A dialysis bag (Visking Company, type 36/32) containing water (20 ml) was immersed in each suspension. After 5 days, the dialysis bags were taken out and the concentrations of the ions in the inner solutions were determined. The suspensions remaining in the vessels were filtered through the Millipore filters, and the filtrates were analysed.

The concentration of pLys was determined by fluorometry at 475 nm with excitation at 335 nm, using *o*-phthalaldehyde and 2-mercaptoethanol according to the method of Roch.[18] The reaction was performed at pH 9.0 by using sodium dihydrogen phosphate–potassium hydroxide buffer. Hereafter, the concentration of pLys is given in terms of molarity of the lysine residue of pLys. The concentration of bromide ion was determined by spectrophotometry at 458 nm, using mercuric thiocyanate and ferric nitrate.[16,17]

The concentrations of calcium and phosphate species in solution were determined by inductively coupled plasma atomic emission spectrometry (ICP–AES) at 393.4 and 214.9 nm, respectively. Working standards were prepared from 1,000 mg/l. standards (Kanto Chemical Co.) by mixing and diluting them with 1% nitric acid. It was confirmed that pLys up to 5m*M* does not interfere with the determination. For the determination of phosphate in the solutions inside the dialysis bags, where the phosphate level was very low (around 0.01m*M*) and pLys was absent, spectrometry at 720 nm using ammonium molybdate and stannous chloride[19] was adopted instead of ICP–AES.

Circular dichroism (CD) spectra of the pLys solutions in the range of 200–250 nm were measured at various pH values on the spectropolarimeter. Solutions of pLys in the three forms (*i.e.*, random coil, α-helix and β-sheet) were prepared according to the method of Greenfield *et al.*[20,21]

RESULTS AND DISCUSSION

Adsorption of pLys on HAp and concurrent release of calcium and phosphate ions from HAp

The adsorbed amounts of pLys and bromide ion (X_{pLys} and X_{Br}) on HAp from aqueous solutions of pLysHBr are shown in Fig. 1 as a function of the equilibrium concentration of pLys (C_{pLys}), where both amounts increased with C_{pLys}, and X_{pLys} was larger than X_{Br}.

It was found that calcium and phosphate ions were released from HAp incongruently (nonconstant in the calcium/phosphorus ratio) during the adsorption of pLys on HAp. The equilibrium concentrations of calcium and phosphate ions in the solutions were plotted against C_{pLys} in Fig. 2(a). The equilibrium concentration of calcium ion (C_{Ca}) increased and that of phosphate ion (C_{Pi}) decreased with C_{pLys}. However, the latter increased after attaining a minimum when C_{pLys} became high. Figure 2(b) shows the relationship between the

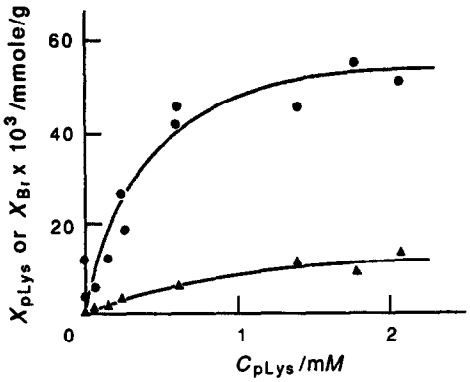

Fig. 1. Adsorbed amounts of pLys and bromide ion on HAp from aqueous solutions of pLysHBr as a function of equilibrium concentration of pLys. No buffering solutions were used in order to avoid the effect of buffering agents on the properties of the adsorbate and the adsorbent. The closed circles and closed triangles correspond to X_{pLys} and X_{Br}, respectively.

Table 1. Equilibrium values (solution volume, concentrations and pH) in adsorption and dissolution experiment with an equilibrium dialysis technique [mean ($n = 4$) ± S.D.]. Initial concentrations: (A) pLys 2.31, Br^- 1.98 m*M*; (B) pLys 3.08, Br^- 2.64 m*M*

	Inside	Outside
(A)		
Equilibrium volume, *ml*	18.7 ± 0.6	31.3 ± 0.6
C_{pLys}		0.155 ± 0.046
C_{Br}	1.01 ± 0.08	1.18 ± 0.02
C_{Ca}	0.354 ± 0.010	0.268 ± 0.120
C_{Pi}	0.0196 ± 0.0099	0.101 ± 0.033
(pH)	6.21 ± 0.05	6.43 ± 0.11
(B)		
Equilibrium volume, *ml*	17.9 ± 1.7	32.1 ± 1.7
C_{pLys}		0.522 ± 0.111
C_{Br}	1.27 ± 0.23	1.62 ± 0.20
C_{Ca}	0.398 ± 0.009	0.301 ± 0.017
C_{Pi}	0.0111 ± 0.0034	0.119 ± 0.006
(pH)	6.44 ± 0.16	6.48 ± 0.05

Concentrations in m*M*.

equilibrium pH of the solutions and C_{pLys}, where the pH had the similar tendency to C_{Pi}.

Migration of released ions through dialyzing membrane

From the outer suspension in contact with the dialysis bag containing water, some of the bromide ions (added as pLysHBr) and some of the calcium and phosphate ions (released from HAp) migrated into the inside of the bag to reach membrane equilibrium. The equilibrium values of the dialysis experiment are listed in Table 1; initial concentration of pLys was 2.31 (A) or 3.08 m*M* (B). The concentrations of anionic species in the outer solution were higher than those in the inner solution, whereas the concentration of calcium ion in the outer solution were lower than that in the inner solution. Equilibrium pH of the outer solution was higher than that of the inner solution.

Effect of pH on adsorption of pLys on HAp

Figure 3(A) shows the adsorbed amounts of pLys and bromide ion on HAp as a function of the equilibrium pH. Initial concentration of pLys was kept constant at 2.89 m*M* and pH was adjusted by addition of sodium hydroxide. The X_{pLys} (closed circles) increased with the pH and then decreased slightly after attaining a

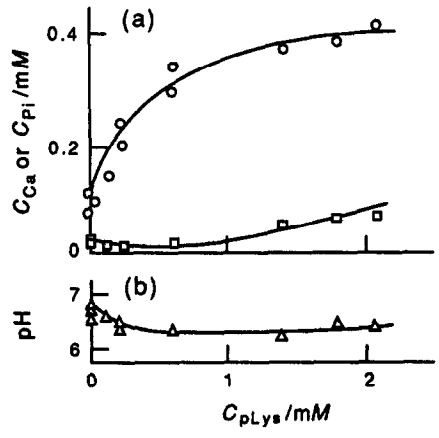

Fig. 2. Relationship between equilibrium concentration of pLys and concentrations of calcium and phosphate ions released from HAp (a), and equilibrium pH of solution (b). The experimental condition was the same as that in Fig. 1. The open circles and open squares correspond to C_{Ca} and C_{Pi}, respectively.

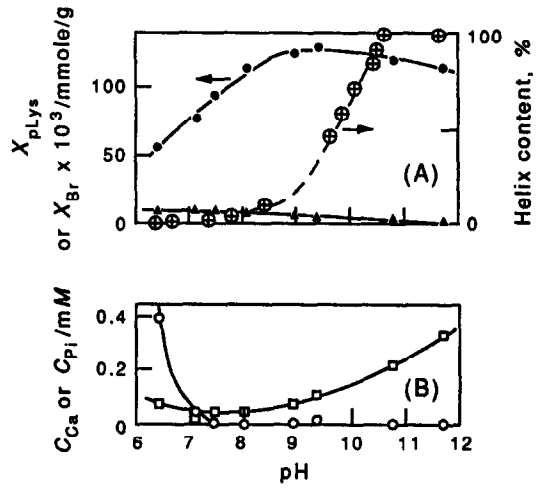

Fig. 3. Relationship between equilibrium pH of solution and adsorbed amounts of pLys and bromide ion on HAp (A), and equilibrium concentrations of calcium and phosphate ions released from HAp (B). All the symbols are the same as those in Figs. 1 and 2. The dotted line on ⊕ in Fig. 3(A) shows the helix content (%) in pLys.

maximum around pH 9.5. On the other hand, X_{Br} (closed triangles) was low and decreased monotonously with the pH. Figure 3(B) shows the equilibrium concentrations of calcium and phosphate ions (C_{Ca} and C_{Pi}) as a function of the equilibrium pH. It shows that C_{Ca} decreased monotonously, and C_{Pi} increased after attaining a minimum with the pH. This dissolution property of HAp in basic media was discussed before.[22]

In aqueous solution of calcium phosphates, the chemical species of Ca^{2+}, PO_4^{3-}, HPO_4^{2-}, $H_2PO_4^-$, H_3PO_4, $CaHPO_4$ and $CaH_2PO_4^+$ are formed through protonation of phosphate ion and through complex formation between calcium and phosphate ions.[23] Moreover, in the presence of pLys, a part of anionic species of phosphate is considered to be bound to amino groups of pLys. The concentration of calcium species determined by the ICP–AES is, therefore, the sum of the concentrations of Ca^{2+}, $CaHPO_4$ and $CaH_2PO_4^+$; the concentration of phosphate species determined simultaneously is the sum of the concentrations of PO_4^{3-}, HPO_4^{2-}, $H_2PO_4^-$, H_3PO_4, $CaHPO_4$, $CaH_2PO_4^+$ and phosphate which is bound to pLys in solution.

The concentrations and activities of chemical species in the solutions inside the dialysis bags can be calculated from the data shown in Table 1 using the first, second and third dissociation constants of phosphoric acid (K_1, K_2 and K_3, respectively) and the formation constants of $CaHPO_4$ and $CaH_2PO_4^+$ (K_x and K_y, respectively).[24] The obtained values were listed in the middle part of Table 2, where brackets and parentheses mean concentration and activity, respectively. The numerical values used were as follows: K_1 7.08×10^{-3}, K_2 6.31×10^{-8}, K_3 4.17×10^{-13},[25] K_x 244 and K_y 8.20.[24] The activity coefficient for each ion (γ) was calculated through the Debye–Hückel equation;

$$\log\gamma = -Az^2I^{\frac{1}{2}}/(1 + Ba^\circ I^{\frac{1}{2}}) \qquad (1)$$

where z is the valency of the ion, I is the ionic strength of the medium, A and B are constants ($A = 0.515$ and $B = 0.329$ at 25° for aqueous solution),[26] and a° is an ion size parameter (H^+ 9, OH^- 3.5, $H_2PO_4^-$ 4.25, HPO_4^{2-} 4, PO_4^{3-} 4, Ca^{2+} 6, Br^- 3 and $CaH_2PO_4^+$ 4.25).[27] The values of $-\log(Ca^{2+})^{10}(PO_4^{3-})^6(OH^-)^2$ were found to be 122.1 and 120.1 for initial concentrations of pLys of 2.31 and 3.08 mM, respectively. These values were in agreement with the literature values of the solubility product (K_{sp}) of HAp (108–125).[28] This result means that the concentrations of calcium and phosphate ions in the inner solutions were limited by each other under the law of the K_{sp} of HAp.

Table 2. Equilibrium values in solutions inside and outside dialysis bag calculated from data listed in Table 1. The symbol Y means the amount of phosphate or bromide ion which is bound to pLys in the solution per lysine residue of pLys: $Y_{Pi} = [Pi]_b^\circ/C_{pLys}$, $Y_{Br} = [Br^-]_b^\circ/C_{pLys}$. Initial concentrations: (A) pLys 2.31, Br^- 1.98 mM; (B) pLys 3.08, Br^- 2.64 mM

	(A)	(B)
$X_{pLys} \times 10^3$, mmole/g	108	126
$X_{Br} \times 10^3$, mmole/g	6.32	7.64
$[H_3PO_4]^i$	1.46×10^{-6}	4.48×10^{-7}
$[H_2PO_4^-]^i$	1.74×10^{-2}	9.13×10^{-2}
$[HPO_4^{2-}]^i$	2.02×10^{-3}	1.82×10^{-3}
$[PO_4^{3-}]^i$	1.68×10^{-9}	2.62×10^{-9}
$[Ca^{2+}]^i$	3.54×10^{-1}	3.98×10^{-1}
$[CaHPO_4]^i$	1.25×10^{-4}	1.23×10^{-4}
$[CaH_2PO_4^+]^i$	4.28×10^{-5}	2.49×10^{-5}
$[H^+]^i$	6.43×10^{-4}	3.80×10^{-4}
$[OH^-]^i$	2.48×10^{-5}	4.23×10^{-5}
$-\log(Ca^{2+})^{10}(PO_4^{3-})^6(OH^-)^2$	122.1	120.1
$[Pi]_f^\circ$	0.0228	0.0130
$[Pi]_b^\circ$	0.0782	0.106
$[Br^-]_f^\circ$	1.16	1.46
$[Br^-]_b^\circ$	0.02	0.16
Y_{Pi}	0.505	0.203
Y_{Br}	0.129	0.307

Concentrations in mM.

When membrane equilibrium is attained, the K_{sp} of HAp are equal on both sides of the membrane, under the assumption that the standard chemical potential of HAp on both sides of the membrane are the same. Therefore, even in the outer solutions containing pLys, the activities of calcium and phosphate ions were mutually restricted under the law of the K_{sp} of HAp. It was concluded that the released phosphate ions in the outer solution exist mainly in the bound state to the amino groups of pLys in the solution. On the other hand, adsorbed pLys expels calcium ion from HAp through the ion-exchange between positively charged amino groups of pLys and calcium ions of HAp to keep electroneutrality of the surface phase. The concomitant increase of C_{Ca} and C_{Pi} shown in Fig. 2(A) were attributed to these facts. The tendency of C_{Pi} to decrease with an increase in C_{Ca} was observed only in the region of low C_{pLys}, where the amount of phosphate ions which were bound to pLys in the solution phase was little.

Taking the Donnan's membrane equilibrium into account, the concentrations of anions, *i.e.*, phosphate and bromide ions, free from both pLys and HAp in the outer solution ($[Pi]_f^\circ$ and $[Br^-]_f^\circ$) can be evaluated from the

following equations;

$$[Pi]_f^o = [H_3PO_4]_f^o + [H_2PO_4^-]_f^o + [HPO_4^{2-}]_f^o + [PO_4^{3-}]_f^o + [CaHPO_4]_f^o + [CaH_2PO_4^+]_f^o \quad (2)$$

$$= [H_3PO_4]^i + [([H_2PO_4^-]^i)^2[Ca^{2+}]^i/[Ca^{2+}]^o]^{\frac{1}{2}} + [HPO_4^{2-}]^i[Ca^{2+}]^i/[Ca^{2+}]^o + [([PO_4^{3-}]^i)^2([Ca^{2+}]^i)^3/([Ca^{2+}]^o)^3]^{\frac{1}{2}} + [CaHPO_4]^i + [CaH_2PO_4^+]^i[H^+]^o/[H^+]^i \quad (2')$$

and

$$[Br^-]_f^o = [([Br^-]^i)^2[Ca^{2+}]^i/[Ca^{2+}]^o]^{\frac{1}{2}} \quad (3)$$

where the subscript "f" means "free", and the superscripts "i" and "o" mean "inside" and "outside", respectively. It was assumed that the activity coefficients for uni- and bivalent ions and the concentrations of neutral species, *i.e.*, H_3PO_4 and $CaHPO_4$, were equal on both sides of the dialysis membrane, and that the cationic species did not interact with pLys. Accordingly, the concentrations of phosphate and bromide ions which were bound to pLys in the outer solution are obtained;

$$[Pi]_b^o = C_{Pi}^o - [Pi]_f^o \quad (4)$$

and

$$[Br^-]_b^o = C_{Br}^o - [Br^-]_f^o \quad (5)$$

The calculated values were listed in the bottom part of Table 2. It shows that phosphate ions in the outer solution exist mostly in the bound state as presumed above.

When the solution pH becomes higher than the PZC of HAp (6.7), the electrostatic attraction between the cationic pLys and the negatively charged HAp surface increases. Thus, X_{pLys} increased with the pH. However, when the pH further increases, pLys comes to lose its positive charges and the conformational change between random coil (at lower pH) and α-helix (at higher pH) occurs in pLys. The dotted line in Fig. 3(A) shows the helix content, estimated from the CD-spectra of the pLys solutions according to the procedure of Greenfield *et al.*[21] It shows that the helix content increased sharply in the range of pH 9–11. However, no pronounced change in X_{pLys} was observed in this region. It was concluded, therefore, that the conformational change of pLys in this region does not give a significant effect on the adsorption of pLys.

In general, as for the adsorption of highly solvated polymers on an impenetrable surface, a fraction of the molecules is actually attached to the sorbent surface, and the remainder dangles in solution.[29] Therefore, the following model explaining the phenomena found in the present study was considered as shown in Fig. 4. A fraction of adsorbed pLys which was actually attached to HAp (c) released calcium ions from HAp (d) by the mechanism of ion-exchange. Co-adsorbed bromide ions were captured by the amino groups of the remaining fraction of adsorbed pLys (b), because the affinity of bromide ion to HAp is very low.[30] In addition to the ion-exchange mechanism, calcium ions were released from HAp through the dissolution of HAp (e). On the other hand, phosphate ions were released only through the dissolution (e). Most of these phosphate ions protonated (g) and/or were bound to pLys in the solution phase (i). Some released phosphate ions formed complexes with calcium ions (h). The dissolution of HAp and the protonation of phosphate ions (g) increased the concentration of hydroxide ion in solution (e) and (f). The similar tendency between the shape of curves of C_{Pi} and (pH) in

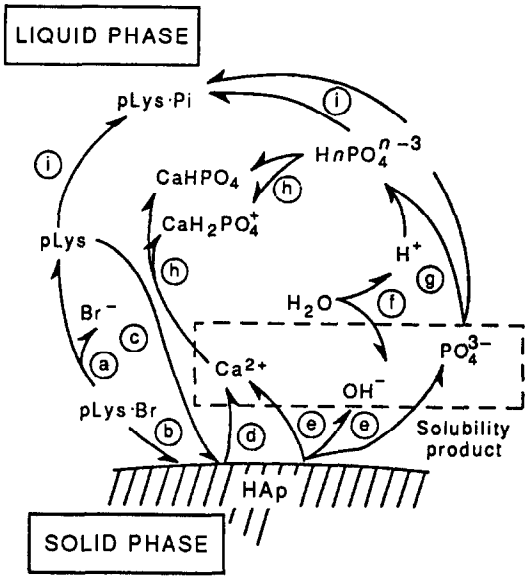

Fig. 4. Scheme of interactions occurring between pLysHBr and HAp. (a) Liberation of bromide ion from pLys, (b) co-adsorption of pLys and bromide ion on HAp, (c) ion-exchange adsorption of pLys on HAp, (d) release of calcium ion through ion-exchange with adsorbed pLys, (e) dissolution of HAp, (f) dissociation of water, (g) protonation of phosphate ion, (h) complex formation between calcium and phosphate ions, (i) counterion binding of phosphate to pLys. For simplicity, arrows are expressed only for the direction of mass transfer or reaction mainly occurs in the course of adsorption of pLys on HAp.

Fig. 2 is attributed to this fact. The concentrations of Ca^{2+}, OH^- and PO_4^{3-} were limited by each other under the restriction of the K_{sp} of HAp (enclosed with the dotted line). Therefore, the release of phosphate ions through the dissolution of HAp (e) was suppressed by the excess release of calcium ions through the ion-exchange (d) when the equilibrium concentration of pLys in solution was low. These facts were responsible for the marked incongruent dissolution properties of HAp in the presence of pLys.

It is known that the constituent ions of HAp having the same sign of electric charge as that of the adsorbed polymeric ions are released from HAp during their adsorption.[31,32] On the other hand, little attention is paid to the behaviour of the constituent ions having a counter-charge to the adsorbate. One of the exceptions is a report of Juriaanse *et al.*[33] They found that not only calcium ions but also phosphate ions were released from HAp in the presence of pLys and poly-L-ornithine; not only phosphate ions but also calcium ions were released in the presence of poly-L-asparaginate. Although they did not refer to it, initial decrease in the released amount of the counter-charge ions, such as that shown in Fig. 2(a), was seen in their reported data. These facts suggest that the dissolution tendency of HAp found in the present study is not specific for the case of adsorption of pLys but is common to those of other cationic peptides, and that the behaviour of phosphate and calcium ions changed place with each other according to the sign of the electric charge of the adsorbed polymeric ions. The results obtained in the present study are, therefore, important as a basis for understanding the interaction between charged polymers and HAp.

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